MIL-R-55444A(EL)
12 February 1970
Superseding MIL-R-55444(EL)
30 November 1965

MILITARY SPECIFICATION

RADIO, SQUAD, RECEIVING AND TRANSMITTING
(UNITS OF)

1. SCOPE

1.1 This specification covers the following units of a small, portable, battery operated, frequency-modulated (fm) squad radio set constructed to provide short range voice communications over a frequency range of 47.00 to 57.00 megahertz (MHz) (See 6.1 and 6.7).

Receiving Set, Radio AN/PRR-9( )
Transmitting Set, Radio AN/PRT-4( )

1.2 The following appendix forms a part of this specification: Reliability Test Plan for Squad Radio.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extend specified herein.

SPECIFICATIONS

FEDERAL

UU-B-36	Bag, Paper, Grocers.
PPP-T-45	Tape, Gummed Paper, Reinforced and Plain, for Sealing and Securing.
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament-Reinforced.
PPP-P-291	Paperboard, Wrapping and Cushioning.
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.
PPP-B-636	Box, Fiberboard.

FSC 5820

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MILITARY

MIL-P-116	Preservation, Methods of.
MIL-T-152	Treatment, Moisture-and-Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	Varnish, Moisture-and-Fungus-Resistant, for Treatment of Communications, Electronic, and Associated Electrical Equipment.
MIL-T-713	Twine and Tape, Lacing and Tying (for use in Electrical and Electronic Equipment).
MIL-M-10578	Metal Conditioner and Rust Remover (Phosphoric Acid Base).
MIL-P-11268	Parts, Materials, and Processes used in Electronic Equipment.
MIL-R-11936	Requirement for and Measurement of Spurious Radiation of 0.55 to 100 MC Radio Transmitter.
MIL-M-13231	Marking of Electronic Items.
MIL-P-13949	Plastic Sheet, Laminated, Copper-Clad (for Printed Wiring).
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-S-19500	Transistors, General Specification for.
MIL-I-46058	Insulating Compound, Electrical.
MIL-P-55110	Printed Wiring Board.

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads, 40 in. X 48 in.; 4-Way Partial and 4-Way Pallets.
MIL-STD-169	Extreme-Temperature Cycle.
MIL-STD-170	Moisture-Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-275	Printed Wiring for Electronic Equipment.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-781	Reliability Tests: Exponential Distribution.
MIL-STD-810	Environmental Test Methods.

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DRAWINGS

ELECTRONICS COMMAND

SC-DL-523470	Transmitting Set, Radio AN/PRT-4( ).
SC-DL-523480	Receiving Set, Radio AN/PRR-9( ).

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Description.-

3.1.1 Receiving Set, Radio AN/PRR-9( ) is a fully transistorized unit contained in a small watertight case constructed to permit attachment to standard combat helmets. Alternately, the unit may be carried in a pocket or attached to combat suspenders. The receiver operates over the frequency range of 47.0 to 57.0 megahertz (MHz) inclusive. Audio presentation is accomplished using an acoustical speaker horn. An auxiliary earphone and cord set are provided for use in cold weather or secure areas. A flexible 18 inch whip antenna is used and one crystal controlled channel is provided for the reception of voice and tone transmissions. Audio squelch is also provided.

3.1.2 Transmitting Set, Radio AN/PRT-4( ) is a fully transistorized unit contained in a small watertight case and constructed for one hand operation. It operates over the frequency range of 47.0 to 57.0 megahertz (MHz) inclusive. Two crystal controlled voice communication-channels, designated Channel 1 and Channel 2, are provided. Channel 1 operates at full power over a usable range of up to one mile. Channel 2 operates at reduced power over a range of up to 500 yards. An ALERT tone signal can be transmitted over either channel to alert the recipient of forthcoming intelligence. Facilities are provided for transmission of a 150 Hz tone simultaneously with the transmission of either voice or the alert tone to permit reception by the Receiver-Transmitter RT-505/PRC-25, and Receiver-Transmitter RT-841/PRC-77, when these receiver-transmitter units are set to receive in the "squelch" mode. A 24-inch telescopic whip antenna is used to provide for efficient storage when not in use.

3.2 Preproduction Samples.- The contractor shall furnish 14 preproduction samples of the squad radio set for approval, provided the Invitation for Bids and the contract require preproduction samples (See 4.3 and 6.2).

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3.3 Construction.- The units of the Squad Radio Set shall be constructed in accordance with the following Drawing and Data Lists:

Transmitting Set, Radio AN/PRT-4( )	SC-DL-523470
Receiving Set, Radio AN/PRR-9( )	SC-DL-523480

In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 Printed Wiring Assemblies.-

3.4.1 Printed Wiring Boards.- Printed wiring boards used in the construction of printed wiring assemblies, shall be products which have passed the requirements of MIL-P-55110, MIL-STD-275 and the master drawing.

3.4.2 Laminate.- The laminate shall be type GB or GH in accordance with MIL-P-13949, unless otherwise specified in the master drawing.

3.4.3 Plating.- Plating of printed wiring boards shall be in accordance with MIL-P-55110 and the master drawing.

3.4.4 Protective Coating.- Printed wiring assemblies shall be protected on both sides with conformal coating. The coating shall be in accordance with MIL-I-46058. For areas where protective coating cannot be applied without impairment of electrical or physical parameters, the contractor shall coordinate with the Contracting Officer's Technical Representative for procedures for masking parts before applying the coating.

3.4.5 Solderability.- Solderability of printed wiring boards shall conform to MIL-P-55110.

3.5 Semiconductors, Transistors, and Diodes.- Unless otherwise specified in equipment drawings, all semiconductors, transistors, and diodes shall conform to MIL-S-19500.

3.6 Cleaning.-

3.6.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, cases, etc.

3.6.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

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3.7 Controls. - All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.8 Finish. - The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.4). The final paint film on Type 1 surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.3).

3.9 Marking. -

3.9.1 General. - Marking shall conform to MIL-M-13231. (See 4.4). Case markings shall be Group 1 as described in that specification.

3.9.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.9.3 Serial Numbers. - Receiving Set, Radio AN/PRR-9( ) and Transmitting Set, Radio AN/PRT-4( ) shall be serial numbered.

3.10 Riveting. - After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.11 Securing of Parts. -

3.11.1 General. - Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.11.2 Securing of Parts by Threaded Fasteners. - Nylok or other screws with plastic devices, and loctite or similar sealants shall not be subjected to temperatures in excess of 250°F. Manufacturers' instructions shall be followed in the application of sealants.

3.12 Soldering. -

3.12.1 Solder. - Solder shall conform to the equipment drawings.

3.12.2 Acid or Acid Salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning of soldering of

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mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.12.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.13 Welding. - Welding shall conform to the equipment drawings.

3.13.1 Cleaning Prior to Welding. - Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.13.2 Process. - Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Insert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium, or stainless steel.

3.13.3 Cleaning after Welding. - Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.14 Wiring and Cabling. - Wiring and cabling shall be neat and sturdy.

3.14.1 Cabling. - Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would

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be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by tape or twine conforming to MIL-T-713. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.14.2 Slack. - Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected, except for certain radio-frequency leads which must be as short as possible for electrical reasons.

(d) To facilitate field repair of broken or cut wires.

(e) To prevent chafing or breaking of individual wires due to repeated flexing of hinged parts.

3.14.3 Protection. - Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.14.4 Support. - Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (1) during and after subjection of the equipment to specified service conditions or (2) after service or repair of the equipment in a normal manner.

3.14.5 Clearance. - Clearance between wires or cables, and parts such as electron tubes and resistors, shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.14.6 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

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3.14.7 Connections.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and non-metallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections, with the exception of the coating specified in 3.4.4.

3.15 Service Conditions.- The equipment shall meet the following service conditions. Where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.15.1 Temperature (see 4.7.1).-

(a) Operating: Ambient temperature in the range of +131°F to -40°F. (The +131°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Non-operating: Exposure in the range of +160°F to -80°F; exposure at high temperature extreme not to exceed 1 hour and at the low temperature extreme not to exceed 24 hours, at any one time.

3.15.2 Moisture Resistance (See 4.7.2).- Exposure to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours for each 24-hour period.

3.15.3 Elevation (See 4.7.3).-

(a) Operating: Up to 10,000 feet above sea level.

(b) Non-operating: Up to 40,000 feet above sea level.

3.15.4 Immersion (See 4.7.4).- Three feet of water for 2 hours.

3.15.5 Orientation.-

(a) Operating (See 4.7.5): The equipment shall be operable in any orientation from normal operating position (that is forward, backward, left, or right).

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(b) Non-operating: Storage in any position for a period of two years. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the preproduction sample (See 3.2) shall be considered as compliance with this requirement.

3.15.6 Sand and Dust (See 4.7.6).- As encountered during heavy traffic over dry clay roads or in desert areas.

3.16 Vibration, Bounce and Shock.-

3.16.1 Vibration, Internal.- The amplitude of vibration of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hz. (See 4.7.7).

3.16.2 Bounce and Shock.- The equipment shall be capable of meeting the requirements of Table I.

TABLE I - Bounce and Shock

Test	Para. No.	Performance After Test
Bounce	4.7.8	Specified performance. (Note A) No physical damage.
Shock, Bench-handling	4.7.9	Specified performance. (Note A) No physical damage.
Shock-drop	4.7.10	Operable. (Note A) Any physical damage shall be minor only. The operational capability shall not be degraded.

NOTE A: The receiver and transmitter shall meet specified performance for the following requirements 3.17.1, 3.17.2, 3.17.3, 3.17.6, 3.18.1, 3.18.2, 3.18.5, 3.18.6, 3.18.19 and 3.18.21.

3.17 Receiver Electrical Requirements- AN/PRR-9( ).-

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3.17.1 Sensitivity (See 4.9.1). - Radio frequency (RF) signal inputs shown below shall produce a signal-plus-noise-to-noise ratio of at least 10 db.

Battery Voltage (VDC)	Signal Input (uv)
6	0.8
5	0.8
4	1.4

3.17.2 Distortion (See 4.9.2). - The overall total harmonic distortion shall not exceed 10 percent when the output power and input voltages are as shown below:

Input Voltage (VDC)	Output Power (mw)
6	5.0
5	5.0
4	2.5

3.17.3 Audio Output Power (See 4.9.3). - The receiver audio output with full volume control shall be capable of supplying 10 milliwatts into a 600 ohm load, with the receiver operating from 5 VDC.

3.17.4 Volume Control (See 4.9.4). - The volume control shall vary the acoustical horn output level. At minimum setting of the control, there shall be no more than 0.10 milliwatt output with no signal input. At the maximum setting of the control, the acoustical horn level shall be not less than the level specified in 3.17.3.

3.17.5 Limiting (See 4.9.5). - The IF limiting characteristic when measured in accordance with 4.9.5 shall be such that the audio output power shall vary not more than ±3 db.

3.17.6 Squelch Sensitivity (See 4.9.6). - The squelch circuitry shall respond to the RF signal input, or less, as shown below:

Battery Voltage (VDC)	Max. Signal Input (uv) -40°F	+86°F	+131°F
6	1.2	1.2	1.2
5	1.2	1.2	1.2
4	2.1	1.7	2.1

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3.17.7 Audio Response (See 4.9.7). - The overall audio response of the receiver shall conform to the table below using the 1000 Hz response as the zero db reference.

Audio Response Limits

Audio Frequency (Hz)	Response (db)
150	-8 or greater
300	-4 ±2
600	0 ±2
1000	0
2500	-4 ±2
5000	-8 or greater

3.17.8 Input Voltage Range. - The receiver shall meet the specified requirements as stated in 3.17.1, 3.17.2, and 3.17.6 when operating from a 4 to 6 volts DC dry battery source. Nominal input voltage is 5.0 volts DC.

3.17.9 Input Power Consumption (See 4.9.8). - The receiver shall not require more than 0.065 watt (13 ma @ 5 volts) of input power when operated at nominal input voltage.

3.17.10 Frequency Range (See 4.9.9). - The receiver shall be capable of operating over the frequency range of 47.0 to 57.0 MHz.

3.17.11 Acoustical Horn Input Impedance (See 4.9.10). - The acoustical horn input circuit shall have an impedance of 600 ohms ±20% and a DC resistance of 220 ohms ±10%.

3.17.12 IF Selectivity (See 4.9.11). - The sealed ceramic ladder filter shall produce the following selectivity curve characteristics:

(a) Bandwidth (6 db points) 40 KHz ±10 KHz.

(b) Bandwidth (60 db points) 72 KHz max.

(c) The ultimate attenuation outside the passband shall be 70 db minimum. The response characteristic in the pass-band shall have no irregularities greater than ± 2 db.

3.17.13 Spurious Responses (See 4.9.12). - There shall be no more than 10 spurious responses which are less than 60 db below the RF signal reference of 1 microvolt, and no more than 2 spurious responses which are less than 40 db below the RF signal reference of 1 microvolt. There shall be no spurious responses less than 35 db below the RF reference signal.

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3.17.14 IF Rejection (See 4.9.13). - The IF rejection ratios shall be greater than 80 db.

3.17.15 Oscillator Radiation (See 4.9.14). - The radiation of either oscillator within the equipment shall produce an output no greater than 100 microvolts at the antenna terminals when the receiver is operating at any frequency within its tuning range.

3.17.16 Image Rejection (See 4.9.15). - The image rejection ratios of the receiver shall be greater than 60 db below a reference level when measured in accordance with 4.9.15.

3.17.17 Discriminator Characteristic (See 4.9.16). - The 455 KHz discriminator shall produce the following characteristics:

(a) Minimum peak separation -40 KHz.

(b) Null shift due to temperature cycling shall be less than ±5 KHz.

(c) The linearity shall be determined at the ±2 volt point either side of the null point. The frequency asymmetry ratio shall not be more than 1.3 to 1.

3.17.18 Audio Distortion at 12 KHz Deviation (See 4.9.17). - The audio response of the receiver shall be in compliance with 3.17.7 when the output is 2.5 times the standard output as stated in 3.17.2 (6 VDC). Under these conditions, the audio distortion shall not be greater than 15%.

3.17.19 Local Oscillator Frequency Accuracy (See 4.9.18). - The local crystal controlled oscillator shall have a frequency accuracy of ±0.008% of the assigned frequency under all service conditions.

3.17.20 10.245 MHz Oscillator Frequency Accuracy (See 4.9.19). - The 10.245 MHz crystal controlled oscillator shall have a frequency accuracy of ±0.008% of the designated frequency under all service conditions.

3.17.21 Frequency Accuracy (See 4.9.20). - The overall frequency accuracy of the receiver shall be within ±4 KHz of the channel frequency.

3.18 Transmitter Electrical Requirements - AN/PRT-4( ). -

3.18.1 Power Output - Channel 1 (See 4.11.1). - The radio frequency power delivered into a standard dummy load shall be not less than 450 milliwatts when operated under standard test conditions (at room ambient). At the temperature extremes of -40°F and +131°F ambient, the radio frequency power output shall be not less than 420 mw.

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3.18.2 Power Output - Channel 2 (See 4.11.2). - The radio frequency power delivered into a standard dummy load shall be not less than 50 milliwatts or more than 250 milliwatts at any Channel 2 frequency assignment which is at or within plus or minus one megahertz from Channel 1 frequency assignment and operating under standard test conditions (at room ambient). At the temperature extremes of -40°F and 131°F ambient, the radio frequency power output shall be not less than 30 milliwatts.

3.18.3 Tone Oscillator Frequency (See 4.11.3). - The frequency of the tone oscillator shall be not less than 600 Hz and not more than 2200 Hz when measured in accordance with 4.11.3.

3.18.4 Modulation Capability (See 4.11.4). - The transmitter RF carrier shall be capable of being deviated in accordance with the following:

Audio Frequency (Hz)	Audio Input (mv)	RF Carrier Deviation (KHz)
300	5.6	8, +5, -3
400	5.6	8, +5, -3
600	5.6	8, +5, -2.5
1000	5.6	8, +5, -2
1600	5.6	8, +5, -2
2500	5.6	8, +5, -2
3200	5.6	8, +5, -2

3.18.5 Catching Range (See 4.11.5). - The catching range of the transmitter shall be not less than 420 KHz nor shall either the plus or minus catching range be less than 180 KHz.

3.18.6 Frequency Accuracy (See 4.11.6). - The frequency of the transmitter shall be within ±4 KHz of the assigned frequency when measured in accordance with 4.11.6.

3.18.7 VCO Temperature Stability (See 4.11.7). - The VCO frequency when tested in accordance with 4.11.7, shall vary not more than ±120 KHz.

3.18.8 10.7 MHz Reference Oscillator Stability (See 4.11.8). - The 10.7 MHz oscillator frequency shall vary not more than ±2.0 KHz when tested in accordance with 4.11.8.

3.18.9 Input Voltage Range (See 4.11.9). - The transmitter shall meet full specification requirements when operating from a DC battery source range of 10.0 to 16.5 volts DC, except the power output at the low input voltage shall be not less than 300 milliwatts in Channel 1 operation (at room ambient). At temperature extremes of -40° and +131°F ambient, the radio frequency power output shall be not less than 275 milliwatts.

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3.18.10 System Efficiency (See 4.11.10). - The efficiency of the transmitter shall be at least 27% when operating under nominal input voltages on Channel 1 at 47 MHz and at least 30% at 51 MHz or 52 MHz and 57 MHz.

3.18.11 Frequency Range (See 4.11.11). - The transmitter shall be capable of operating over the frequency range of 47.0 to 57.0 megahertz.

3.18.12 Audio Input Impedance (See 4.11.12). - This circuit shall have an input impedance of not less than 800 ohms and not more than 3000 ohms at frequencies in the range of 300 to 3000 Hz.

3.18.13 Deviation Limiting (See 4.11.13). - Deviation limiting shall occur at a maximum of ±19 KHz deviation with an audio input level increase of no more than 20 db above the level prescribed for nominal ±8 KHz deviation as referred to in 3.18.4.

3.18.14 Spurious Radiations (See 4.11.14). - In the 20 to 400 megahertz range, there shall be no more than seven spurious radiations, except for harmonics. The maximum number of spurious and their suppression levels are shown in the table below:

Suppression in DB	Number of Spurious
70 - 79	4
60 - 69	5
50 - 59	3
40 - 49	3

3.18.15 Harmonic Radiations (See 4.11.15). - Harmonics of the 47-57 megahertz radiated frequencies shall be suppressed at least 35 db below the fundamental level.

3.18.16 AM Modulation (See 4.11.16). - Amplitude modulation of the RF carrier shall be one percent or less when measured in accordance with 4.11.16.

3.18.17 Antenna Loading (See 4.11.17). - The antenna loading coil shall have sufficient range to accomplish maximum loading of the 24 inch whip antenna to the final stage of the transmitter at any frequency assignment in the 47-57 megahertz range.

3.18.18 Audio Frequency Response (See 4.11.18). - The audio frequency response characteristic shall increase 2.5 ±1.5 db between 200 Hz and 800 Hz and shall not vary more than ±1.5 db between 800 Hz and 3500 Hz.

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3.18.19 System Distortion (See 4.11.19 and 4.11.24). - The amplitude distortion in the output of the tone squelch generator shall not exceed 10 percent. The overall total harmonic distortion shall not exceed 10 percent.

3.18.20 Regulated Voltage (See 4.11.20). -

3.18.20.1 The regulated voltage shall be not less than 7.6 VDC and not more than 8.2 VDC under standard conditions.

3.18.20.2 The regulated voltage with input voltage variations from 10 VDC to 16.5 VDC shall be within +0.05 VDC and -0.10 VDC of the nominal regulated voltage measured under standard conditions.

3.18.20.3 At the temperature extremes of -40°F and +131°F ambient, the regulated voltage shall not vary more than ±0.15 VDC from the nominal regulated voltage measured under standard test conditions.

3.18.21 Tone Squelch Generator Frequency (See 4.11.21). - The frequency of the tone squelch generator shall be 150 ±2.0 Hz when measured in accordance with 4.11.21.

3.18.22 Tone Squelch Generator Modulation (See 4.11.22). - The tone squelch generator shall modulate (deviate) the transmitter frequency 3.0 +1.5 KHz, -0.5 KHz.

3.18.23 Tone Squelch Generator Temperature Stability (See 4.11.23). The frequency of the tone squelch generator when tested in accordance with 4.11.23 shall not vary more than ±4.0 Hz.

3.19 Interchangeability (See 4.12).- Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.20 Reliability Test (See 4.13).- Each receiver and transmitter undergoing reliability testing shall meet the requirements of the Reliability Test Plan for Squad Radio (Appendix A) after being subjected to the tests specified therein.

3.21 Technical Literature.- Technical literature shall be furnished as specified in the contract. (See 6.2).

3.22 Preconditioning.-

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3.22.1 Bounce Preconditioning.- The units shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1.

3.22.2 Electrical Preconditioning.- Each receiver and transmitter shall be electrically preconditioned for 24 hours without degradation of sensitivity or power output. (See 4.6.2).

3.23 Operational Inspection.- The operational test specified in 4.14 shall be performed to determine that the equipment is operable prior to packaging and at such other times as required.

3.24 Tropicalization.-

3.24.1 Treatment of Materials.- Fungus-inert materials shall be used where practicable. These materials shall be in accordance with requirement 4 of MIL-STD-454. When materials are required to be treated, they shall be cleaned in accordance with requirement 9 of MIL-STD-454 and then shall be treated in accordance with 3.24.2 through 3.24.5 and as follows:

Cotton Thread and Duck (See MIL-P-11268)
Cork (See MIL-P-11268)
Felt, Hair or Wool (See MIL-P-11268)
Leather (See MIL-P-11268)
Linen Thread (See MIL-P-11268)
Paper, Paperboard, Cardboard, Organic Fiberboard, Vulcanized Fiber, etc. (See MIL-P-11268)
Thermosetting Plastic Materials Using Cotton, Linen, or Wood-Flour as a Filler or Base (See MIL-P-11268)

3.24.2 Treating Compounds.- Treating compounds containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating compound is compatible with the material or surface to be treated. Selection of treating compound shall be such that there shall be no increase in flammability of the treated material.

3.24.3 Toxicity.- Treating compounds shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.24.4 Flexibility.- Treatment shall not affect the flexibility of the treated material, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

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3.24.5 Statement of Treatment.- The statement of treatment shall indicate whether or not nutrient materials have been employed. If nutrient materials have been used, it shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposes to use. (See 6.2).

3.24.6 Over-all Treatment (Moisture and Fungus Proofing Treatment (MFP)).- When over-all tropicalization treatment of equipment is a contractual requirement, a moisture and fungus resistant coating of varnish conforming to MIL-V-173 shall be applied in accordance with MIL-T-152. The moisture-fungus resistant coating shall be applied after the equipment has been cleaned to remove such contaminants as corrosion products, dust, fingerprints, lubricating oils, mold release agents, sand, and solder fluxes. When tape is used on parts of equipment that are masked against treatments, the tape shall be an electrical grade. In the application of the MFP coating, care shall be taken to assure that the coating does not affect circuit constants and thereby degrade performance.

3.24.7 Fungus (See 4.18).- The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.18. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.25 Salt Fog (See 4.19).- The equipment and assemblies shall be exposed in their operating configuration, to 48 hours of salt fog in accordance with 4.19 and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.26 Workmanship.- The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of 3.3 through 3.14 herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of Inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (See 4.3).

(b) Inspection covered by subsidiary documents (See 4.4).

(c) Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery (See 4.5).

(2) Quality conformance inspection of preparation for delivery (See 4.15).

4.3 Preproduction Inspection. - This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, the inspection specified for Group A, Group B and Group C (see Tables II, IV, and V, respectively) and the inspection specified in 4.13. The preproduction inspection shall be performed in the order shown in Table II according to the number of samples specified in the contract or order.

TABLE II

Order of Preproduction Inspection

Inspection*	Unit 1	Unit 2	Unit 3	Unit 4	Units 5-14	Req Para	Insp Para
Non-Damaging							
Temperature	1					3.15.1	4.7.1
Elevation		1				3.15.3	4.7.3
Immersion		2				3.15.4	4.7.4
Sand and Dust		3				3.15.6	4.7.6
Vibration		4				3.16.1	4.7.7

*NOTE 1: Other preproduction inspection may precede, follow, or be interspersed between the non-damaging tests.

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Inspection *	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5-14	Req Para	Insp Para
Potentially Damaging							
Bounce		5				3.16.2	4.7.8
Shock, Bench-handling		6				3.16.2	4.7.9
Shock, Drop		7				3.16.2	4.7.10
Moisture Resistance	2					3.15.2	4.7.2
Orientation	3					3.15.5	4.7.5
Fungus			1**			3.24.7	4.18
Salt Fog				1**		3.25	4.19
Reliability Test					1	3.20	4.13

** NOTE 2: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.4 Inspection Covered by Subsidiary Documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where Required
Printed Wiring Assemblies	3.4
Semiconductors; Transistors, and Diodes	3.5
Finish, Protective	3.8
Marking	3.9
Workmanship	3.26

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery.- Each unit of each lot of equipment shall be subjected to bounce and electrical preconditioning (See 3.22.1 and 3.22.2) prior to inspection. The supplier shall perform the inspection specified in 4.4, 4.5.1 through 4.5.3.2, and 4.13. This does not relieve the supplier of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce or electrical preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the supplier's inspection procedures and examine the supplier's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (See 6.8). The test equipment for Government verification inspection shall be made available by the supplier.

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4.5.1 Group A Inspection.- Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table III. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105, and the AQL's indicated in Table III. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.14) shall be last. (See 6.6).

TABLE III - Group A Inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical	3.26	4.17	1.0%	4.0%
Receiver				
Printed Wiring Board				
Acoustical Earhorn				
Antenna				
Auxiliary Earphone				
Assembled Receiver				
Transmitter				
Printed Wiring Boards				
Antenna				
Battery Case				
Assembled Transmitter				
Electrical				
Receiver				
Sensitivity	3.17.1	4.9.1	1.5% for the group	not applicable (N/A)
Distortion	3.17.2	4.9.2		
Audio Output Power	3.17.3	4.9.3		
Volume Control	3.17.4	4.9.4		
Limiting	3.17.5	4.9.5		
Squelch Sensitivity	3.17.6	4.9.6		

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Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Transmitter				
Power Output (Channel 1)	3.18.1	4.11.1	1.5% for the group	N/A
Power Output (Channel 2)	3.18.2	4.11.2		
Modulation Capability	3.18.4	4.11.4		
Catching Range	3.18.5	4.11.5		
System Distortion	3.18.19	4.11.19		
Tone Squelch Generator Frequency	3.18.21	4.11.21		
Tone Squelch Generator Modulation	3.18.22	4.11.22		
Operational Inspection	3.23	4.14	1%	N/A

4.5.2 Group B Inspection. - This inspection (including sampling) shall conform to Table IV and to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met Group A inspection.

TABLE IV - Group B Inspection

Inspection	Req Para	Insp Para	AQL	Level
Receiver				
Audio Response	3.17.7	4.9.7	4.0% for the group	S-4
Input Voltage Range	3.17.8			
Input Power Consumption	3.17.9	4.9.8		
Audio Distortion at 12 KHz Deviation	3.17.18	4.9.17		
Local Oscillator Frequency Accuracy	3.17.19	4.9.18		
10.245 MHz Oscillator Frequency Accuracy	3.17.20	4.9.19		
Transmitter				
Frequency Accuracy	3.18.6	4.11.6	4.0% for the group	S-4
Input Voltage Range	3.18.9	4.11.9		
System Efficiency	3.18.10	4.11.10		
Regulated Voltage	3.18.20	4.11.20		
Interchangeability	3.19	4.12	6.5%	S-4

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4.5.2.1 Order of Inspection Within Group B. - Group B inspection shall be performed in any order which is satisfactory to the government.

4.5.3 Group C Inspection. - This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. Each preproduction sample unit used for the salt fog or fungus test need meet only those environmental Group C inspections indicated for that unit in Table II. Sample units shall be selected in accordance with Paragraph 4.5.3.1.

TABLE V - Group C Inspection

Subgroup I (See 4.5.3.1.1)

Inspection	Req Para	Insp Para
Receiver:		
Acoustical Horn Input Impedance	3.17.11	4.9.10
IF Selectivity	3.17.12	4.9.11
Spurious Response	3.17.13	4.9.12
IF Rejection	3.17.14	4.9.13
Oscillator Radiation	3.17.15	4.9.14
Image Rejection	3.17.16	4.9.15
Discriminator Characteristic	3.17.17	4.9.16
Frequency Accuracy	3.17.21	4.9.20
Transmitter:		
Tone Oscillator Frequency	3.18.3	4.11.3
Modulation Capability	3.18.4	4.11.4
Audio Input Impedance	3.18.12	4.11.12
Deviation Limiting	3.18.13	4.11.13
Spurious Radiations	3.18.14	4.11.14
Harmonic Radiations	3.18.15	4.11.15
AM Modulation	3.18.16	4.11.16
Antenna Loading	3.18.17	4.11.17
Audio Frequency Response	3.18.18	4.11.18

Subgroup II (See 4.5.3.1.1)

Receiver:		
Frequency Range	3.17.10	4.9.9
Transmitter:		
VFO Temperature Stability	3.18.7	4.11.7
10.7 MHz Reference Oscillator Stability	3.18.8	4.11.8
Tone Squelch Gen Temp Stability	3.18.23	4.11.23
Frequency Range	3.18.11	4.11.11

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Subgroup III (See 4.5.3.1.2)

Inspection	Req Para	Insp Para
Receiver (Assembled):		
Immersion	3.15.4	4.7.4
Transmitter (Assembled):		
Immersion	3.15.4	4.7.4

Subgroup IV (See 4.5.3.1.3)

Receiver:		
Temperature	3.15.1	4.7.1
Vibration, Internal	3.16.1	4.7.7
Bounce	3.16.2	4.7.8
Shock, Drop	3.16.2	4.7.10
Moisture Resistance	3.15.2	4.7.2
Transmitter:		
Temperature	3.15.1	4.7.1
Vibration, Internal	3.16.1	4.7.7
Bounce	3.16.2	4.7.8
Shock, Drop	3.16.2	4.7.10
Moisture Resistance	3.15.2	4.7.2

Subgroup V (See 4.5.3.1.4)

Receiver:		
Elevation	3.15.3	4.7.3
Orientation	3.15.5	4.7.5
Sand and Dust	3.15.6	4.7.6
Shock, Bench-handling	3.16.2	4.7.9
Fungus	3.24.7	*4.18
Salt Fog	3.25	*4.19
Transmitter:		
Elevation	3.15.3	4.7.3
Orientation	3.15.5	4.7.5
Sand and Dust	3.15.6	4.7.6
Shock, Bench-handling	3.16.2	4.7.9
Fungus	3.24.7	*4.18
Salt Fog	3.25	*4.19

*This test shall be performed on a separate sample.

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4.5.3.1 Sampling for Inspection of Equipment. - Units (Each unit shall consist of one receiver and one transmitter.) selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.3.1.1 Subgroups I and II. - For this subgroup, two units from the first 50 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units from each successive 100, or two per month if less than 100 are produced monthly, shall be selected.

4.5.3.1.2 Subgroup III. - For this subgroup, two units from the first 50 production units shall be selected for the Group C inspection. For subsequent Group C inspections, two units from each successive 250, or two per month, if less than 250 are produced monthly, shall be selected.

4.5.3.1.3 Subgroup IV. - For this subgroup, two units from the first 50 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units shall be selected from each successive 500, or two every two months, if less than 500 are produced during this two month period.

4.5.3.1.4 Subgroup V. - For this subgroup, four units from the first 50 production units shall be selected for the first Group C inspection. No subsequent Group C inspection for the tests in this subgroup shall be required.

4.5.3.2 Noncompliance. - The supplier shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The supplier shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspection which the sample failed at the option of the government.) and Groups A and B inspections may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (See 6.5)

4.5.4 Reinspection of Conforming Group C Sampling Units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

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4.6 Preconditioning.-

4.6.1 Bounce Preconditioning.- The receiver and transmitter shall be placed in its normal operating position on the table of the package tester, type 1000-SC, as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment to a distance of three inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to acceptance inspection. (See 3.22.1).

4.6.2 Electrical Preconditioning.- The receiver and transmitter shall be operated for a period of 24 hours under the conditions of Test Level "E" of MIL-STD-781. Standard test conditions shall apply, except as modified by Test Level "E" of MIL-STD-781. The receiver and transmitter shall be tested for compliance with 3.17.1 and 3.18.1 before and after the preconditioning period. (See 3.22.2).

4.7 Service Condition Tests.-

4.7.1 Temperature Test for Equipment.- The units shall be subjected to the temperature cycle shown in MIL-STD-169. For the receiver and transmitter the following measurements shall be made at steps 1, 4 (131°F), 8, and 10.

Measurement	Paragraph
Sensitivity	4.9.1
Distortion	4.9.2
Audio Output Power	4.9.3
Squelch Sensitivity	4.9.6
Catching Range	4.11.5
Power Ourput	4.11.1 and 4.11.2
Tone Oscillator Frequency	4.11.3
System Distortion	4.11.19
Modulation Capability	4.11.4
Tone Squelch Generator Frequency	4.11.21
Tone Squelch Generator Modulation	4.11.22

4.7.2 Moisture Resistance.- The units shall be tested as follows:

4.7.2.1 Test Conditions.-

(a) Do not remove equipment from the humidity chamber for measurement

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

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(c) The receiver and transmitter shall be tested with the seals intact for three 48 hour cycles. Bring out leads for electrical measurements through holes in the equipment enclosure and seal against the entrance of moisture. The last two cycles will be performed with cases removed.

4.7.2.2 Test Procedure. - The equipment shall be tested as follows.

(a) Dry at 130 ±5° for 24 hours.

(b) Condition at 77° ±5° and 40 to 50 percent relative humidity for 24 hours.

(c) Take the measurements if applicable, as specified in 4.7.2.3 and readjust, if necessary, to meet specification requirements.

(d) All units shall be subjected to a total of five continuous 48 hour cycles. Temperature, relative humidity and period of time for each portion of the cycle shall conform to MIL-STD-170. The measurements specified in 4.7.2.3 shall be made during the periods shown on the standard and shall comply with 4.7.2.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5° at 40 to 60 percent relative humidity. Then adjust for optimum performance using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.7.2.3.

4.7.2.3 Performance. - There shall be no mechanical deterioration as evidenced by corrosion of metal parts or binding of rotating parts. The following measurements shall be made during the test.

(a) Sensitivity (4.9.1).

(b) Distortion (4.9.2).

(c) Audio Output Power (4.9.3).

(d) Squelch Sensitivity (4.9.6).

(e) Catching Range (4.11.5).

(f) Power Output (4.11.1 and 4.11.2).

(g) Tone Oscillator Frequency (4.11.3).

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(h) Modulation Capability (4.11.4).

(i) Tone Squelch Generator Frequency (4.11.21).

(j) Tone Squelch Generator Modulation (4.11.22).

4.7.2.4 Failure. - If the unit fails to meet the performance specified in 4.7.2.3 during cycling, it shall be realigned or readjusted once. If the equipment then fails to meet specified performance, or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, it does not pass the test.

4.7.3 Elevation Test. - The unit, placed in its normal operating position in an altitude chamber, shall be tested as follows.

(a) With the unit placed in the chamber, operating and stabilized at room temperature, perform the following tests:

Paragraph	Test
4.9.3	Receiver Audio Output Power
4.11.4	Transmitter Modulation Capability

(b) Reduce the barometric pressure to 20.6 ±0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the test specified in 4.7.3(a). The unit shall meet specified performance for this test.

(c) Remove the power from the equipment and further reduce the barometric pressure to 5.5 ±0.1 inches of mercury (simulated 40,000 feet above sea level). Maintain this condition for one hour.

(d) Return the chamber to ambient pressure. Apply power to the unit and after stabilization, repeat the test specified in 4.7.3(a). The equipment shall meet specified performance for this test.

4.7.4 Immersion. - The receiver and transmitter shall be immersed to a minimum depth of 3 feet of fresh water for a period of 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F or more above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within 2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2 hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence

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of leakage. The following items external to the watertight compartments shall be removed prior to the immersion test: (1) Transmitter Microphone, Loading Coil Housing and Antenna; (2) Receiver Acoustical Earhorn, Loading Coil Housing and Antenna.

4.7.5 Orientation Test. - The equipment shall meet the operational test of 4.14 independent of orientation.

4.7.6 Sand and Dust Test. - The equipment shall be tested as follows.

(a) Place the equipment in its normal operating position within the test chamber. Raise and maintain the sand and dust density at 0.1 to 0.5 gram per cubic foot within the test space. The relative humidity shall not exceed 30 percent at any time during the test.

(b) The sand and dust shall be of angular structure and shall have a particle size such that the particles can pass through screens as follows:

Screen, US Standard Sieve Series	Percent
100 - mesh	100
140 - mesh	98 ±2
200 - mesh	90 ±2
325 - mesh	75 ±2

(c) Chemical analysis of the sand and dust shall be as follows:

Substance	Percent by Weight
SiO_2	97 to 99
Fe_2O_3	0 to 2
Al_2O_3	0 to 1
TiO_2	0 to 2
MgO	0 to 1
Ignition Losses	0 to 2

(d) Maintain internal temperature of test chamber at 77° ±4° for a period of 6 hours, with sand and dust velocity through the test chamber between 100 and 500 feet per minute.

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(e) After 6 hours of above conditions, raise the temperature to 160° ±4°F and maintain it at this value for 6 hours. The other conditions shall remain the same.

(f) Remove equipment from test chamber and allow it to cool to room temperature. Accumulated dust shall be removed by brushing, wiping, or shaking, with care taken to avoid introduction of additional dust into the equipment. However, dust shall not be removed by either blast or vacuum cleaning. The receiver and transmitter shall meet the sensitivity (4.9.1), power output (4.11.1), and operational (4.14) tests at 47 MHz.

(g) Open the equipment. There shall be no evidence of deterioration of either external or internal parts. Abrasion of external finish is permissible.

4.7.7 Test for Internal Vibration. - Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.16.1.

(a) Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 Hz. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of three directions, change the frequency in steps of one Hz and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

(e) At the completion of the test, the receiver and transmitter shall meet the sensitivity (4.9.1) and power output (4.11.1) tests respectively.

4.7.8 Bounce Test. - The equipment in its case shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The test shall be as follows.

(a) Cover the tester bed with a panel of 1/2 inch plywood. Space six penny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

(b) Place the equipment on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

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(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ±2 rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

(d) The equipment shall be tested at the end of each 1/2 hour period for sensitivity (4.9.1) and power output (4.11.1). At the completion of the bounce test, the equipment shall be tested for compliance with 3.16.2.

4.7.9 Shock Test, Bench-Handling. - The receiver and transmitter chassis and front panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid 2 inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing.

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of 4 drops.

(b) Upon completion, the receiver and transmitter shall be tested for compliance with 3.16.2.

4.7.10 Shock Test, Drop. -

(a) Test Method. - The receiver and transmitter shall be dropped a distance of four feet on each corner, edge and face (total of 26 drops). Following the 26 drops, the receiver shall be dropped an additional 10 times while mounted to the combat helmet and liner. No catastrophic failures shall result from this test, except that crystal units may be replaced without penalty. Drops shall be made from a quick-release hook or pendulum tester as made by L.A.B. Corporation, Skaneateles, New York, or equal. The floor or barrier receiving the impact shall be of 2-inch fir wood backed by concrete or a rigid steel frame. Except for abrasion, paint chipping and small local deformations, no damage shall occur.

(b) Performance. - Upon completion of the test and before repairs are made, the receiver and transmitter shall be tested for compliance with 3.16.2.

4.8 Standard Test Conditions, Receiving. - All tests shall be conducted in an electromagnetically screened room conforming to MIL-A-11936, under the following conditions, unless otherwise specified for the individual tests.

Temperature	Room Ambient
Humidity	Room Ambient
Dummy Antenna (for tests through Loading Coil)	39 Ohm Resistor and 5 pf Capacitor in series with 50-ohm Signal Generator

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Input Voltage	5.0 VDC Nominal
RF Input	1.0 Microvolts rms, 8 KHz Deviation at 1000 Hz Modulation Frequency
Squelch	Disabled
Audio Output Level	5 Milliwatts
Audio Output Load Impedance	600 ohms ±10 Percent (AC) and 220 ohms ±10% (DC)
Frequency	51.00 MHz

4.9 Receiver Electrical Tests - AN/PRR-9( ). -

4.9.1 Sensitivity (See 3.17.1). - The receiver shall be operated under standard test conditions as stated in 3.17.1 with the input signal modulation set at 8 KHz deviation at 1000 Hz. The receiver output shall be set for 5 milliwatts at 6 and 5 volts. At 4 volts input, the receiver output shall be set at 2.5 milliwatts. Measurements shall be made at 47, 51 and 57 MHz.

4.9.1.1 Test Procedures. -

(a) Replace the whip antenna with the special AN/PRR-9( ) antenna pad.

(b) Connect a Measurement Corporation Signal Generator, Model 560, or equivalent, to the special pad.

(c) The generator frequency shall be adjusted to receiver channel frequency.

(d) The generator output level shall be adjusted to the specified limit (See 3.17.1).

(e) Connect a General Radio Output Power Meter, Model 583A, or equivalent, to the receiver phone jack. Use 600 ohm load.

(f) Connect a Hewlett-Packard Vacuum Tube Voltmeter, Model 400D, or equivalent, to the power meter terminals.

(g) Use an adjustable DC Power Supply, Hewlett-Packard Model 721A, or equivalent to power the receiver under test.

(h) Disable the receiver squelch by rotating volume control to extreme clockwise position.

(i) Adjust receiver audio output to the level specified in 4.9.1.

(j) The signal-plus-noise-to-noise-ratio, in decibels, obtained for each of the supply voltages shall not be less than the level specified in 3.17.1.

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4.9.2 Distortion (See 3.17.2). - The receiver shall be operated under standard test conditions except that the input RF level of 5000 microvolts shall be modulated by 1000 Hz for 8 KHz deviation. The receiver output power shall be fed into a Hewlett-Packard Distortion Analyzer, Model 330D or equal. Distortion measurements shall be recorded to determine compliance with 3.17.2.

4.9.2.1 Test Procedures. -

(a) The test procedures shall be the same as 4.9.1.1 except that 4.9.1.1(d) and 4.9.1.1(j) shall not apply.

(b) Connect a Hewlett-Packard Model 330D Distortion Analyzer, or equivalent, to the output power meter terminals.

(c) The generator output level shall be adjusted to 5000 microvolts.

(d) Total harmonic distortion obtained at each of the supply voltages shall not be greater than the level specified in 3.17.2.

4.9.3 Audio Output Power (See 3.17.3). - The receiver power output operating at 6 VDC under standard test conditions shall be fed into a General Radio Output Power Meter, Type 583-A or equal to determine compliance with 3.17.3.

4.9.3.1 Test Procedures. -

(a) The test procedures shall be the same as 4.9.1.1 except that 4.9.1.1(d), 4.9.1.1(i), and 4.9.1.1(j) shall not apply.

(b) The generator output shall be adjusted to 5000 microvolts.

(c) Set DC Power Supply for 5.0 volt operation.

(d) With receiver volume control set at extreme clockwise position, the audio power output shall be no less than the level specified in 3.17.3.

4.9.4 Volume Control (See 3.17.4). - The receiver output shall be fed into the power meter (See 4.9.3) and measurements made to determine compliance with 3.17.4.

4.9.4.1 Test Procedures. -

(a) The test procedure shall be the same as 4.9.1.1 except that 4.9.1.1(b), 4.9.1.1(c), 4.9.1.1(d), 4.9.1.1(i), and 4.9.1.1(j) shall not apply.

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(b) At minimum setting of volume control, power output level shall be no greater than level specified in 3.17.4.

4.9.5 Limiting (See 3.17.5). - The receiver shall be operated under standard test conditions. The input RF signal shall be modulated at 1000 Hz and deviation set at 8 KHz. The receiver output reference shall be set at 5 milliwatts output with the volume control and with the pad input RF signal level at 100,000 microvolts. Successive measurements shall then be made according to the list below. The power output variations shall be noted to determine compliance with 3.17.5.

Input Voltage	Pad Input (uv)
5 VDC	100,000
	10,000
	1,000
	500
	100
	10
	1.0

4.9.5.1 Test Procedures. -

(a) The test procedures shall be the same as 4.9.1.1 except that 4.9.1.1(d) and 4.9.1.1(j) shall not apply.

(b) The generator output level shall be adjusted to the specified limit (See 4.9.5).

(c) Decrease generator output slowly to 1 microvolt and note any changes in audio output. Changes in audio output power shall be no greater than limit specified in 3.17.5.

4.9.6 Squelch Sensitivity (See 3.17.6). - With supply voltage and generator output level adjusted for limits specified in 3.17.6, there shall be audio output indicated on the output power meter. When the RF signal is removed from the receiver input, the audio output shall be zero.

4.9.6.1 Test Procedures. -

(a) The test procedures shall be the same as 4.9.1.1 except that 4.9.1.1(d), 4.9.1.1(h), and 4.9.1.1(j) shall not apply.

(b) Turn squelch on by first turning volume control to extreme counterclockwise position. Turn volume control clockwise to mid-position.

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(c) The generator output level shall be adjusted to the specified limit (See 3.17.6).

(d) The quelch sensitivity for each of the supply voltages shall be no greater than the levels specified in 3.17.6. When the RF signal is removed from the receiver input, the audio output shall be zero.

4.9.7 Audio Response (See 3.17.7). - The overall receiver response shall be made with the input RF signal modulation set at 8KHz deviation and the signal level set at 5000 uv. The output power shall be set at 5 milliwatts when 1000 Hz modulation is applied and the input voltage is at 5 volts. The receiver output response shall be measured to determine compliance with 3.17.7.

4.9.7.1 Test Procedures. -

(a) The test procedures shall be the same as 4.9.1.1 except that 4.9.1.1(d) and 4.9.1.1(j) shall not apply.

(b) The RF signal generator shall be modulated with an external audio oscillator, Hewlett-Packard Model 200CD, or equivalent.

(c) The generator output level shall be adjusted to the specified limit (See 4.9.7).

(d) The overall audio response of the receiver shall be within limits specified in 3.17.7.

4.9.8 Input Power Consumption (See 3.17.9). - The input power to the receiver under nominal input voltage conditions shall be measured to determine compliance with 3.17.9.

4.9.8.1 Test Procedures. -

(a) Connect a General Radio Output Power Meter, Model 583A, or equivalent, to the receiver phone jack. Use 600 ohm load.

(b) Use an adjustable DC Power Supply, Hewlett-Packard Model 721A, or equivalent, to power the receiver under test.

(c) Use a Hewlett-Packard clip on milliammeter, Model 428A, or equivalent, to measure receiver current.

(d) Turn squelch on by first turning volume control to extreme counter-clockwise position, then turn volume control clockwise to mid-position.

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(e) With receiver squelched and DC supply adjusted for 5.0 volt operation, current measured on milliammeter shall be no greater than limit specified in 3.17.9.

4.9.9 Frequency Range (See 3.17.10). - The receiver shall be operated under standard test conditions. To determine compliance with 3.17.10, the tests listed below shall be conducted.

(a) Sensitivity (4.9.1).

(b) Audio Power Output (4.9.3).

4.9.10 Acoustical Horn Input Impedance (See 3.17.11). - The input impedance and DC resistance shall be measured to determine compliance with 3.17.11, by connecting an acoustical horn under test to a General Radio impedance bridge, Model 1650-A, or equivalent.

4.9.11 IF Selectivity (See 3.17.12). - The IF shall be monitored with a high impedance meter or ACVTVM. Adjust the RF input to produce a reference IF level of 4.8mv. To measure the 6db points, increase the level by 6db, vary the signal frequency until the ACVTVM returns to reference level. To determine the 60db points of the system, return the RF input to the center of the passband and increase the level by 60db. The point at which the meter again indicates the reference IF level will determine the bandwidth at 60db.

4.9.11.1 Test Procedures. -

(a) The test procedures shall be the same as 4.9.11 except that 4.9.1.1(d) and 4.9.1.1(j) shall not apply.

(b) Connect Hewlett-Packard Vacuum Tube Voltmeter, Model 400D, or equivalent, to output of ceramic ladder filter TP-2.

(c) Adjust RF signal generator output to level specified in 4.9.11.

(d) Measure generator frequency with a Hewlett-Packard Model 524D Electronic Counter, or equivalent.

(e) The differences in frequencies measured at 6 and 60db levels shall be within limits specified in 3.17.12.

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4.9.12 Spurious Response (See 3.17.13). - The receiver shall be operated under standard test conditions. The amount of quieting produced by an unmodulated RF carrier of 1 microvolt shall be used as a reference. The signal generator output shall be increased by 60 db and the generator tuned throughout the range of 20 to 600 MHz. Those frequencies which produce quieting equal to or greater than the reference shall be recorded to determine compliance with 3.17.13. This test shall be made at the following frequencies: 47.00, 51.00, and 57.00 MHz.

4.9.12.1 Test Procedures. -

(a) The test procedures shall be the same as 4.9.1.1 except that 4.9.1.1(i) and 4.9.1.1(j) shall not apply.

(b) The amount of quieting produced by an unmodulated RF carrier of 1 microvolt shall be used as a reference.

(c) The signal generator output shall be increased by 60 db and the generator tuned through the range of 20 to 400 MHz.

(d) Spurious responses which produce quieting equal to or greater than the reference shall be measured with a Hewlett-Packard Counter, Model 524, or equivalent.

(e) Spurious responses measured shall be within limits specified in 3.17.13.

4.9.13 IF Rejection (See 3.17.14). - The receiver shall be tuned to 51 megahertz. The amount of quieting produced by a 1 microvolt carrier signal shall be used as a reference. The signal generator shall then be tuned to 455KHz and 10.7MHz and its output raised until the amount of quieting is equal to the reference. The number of microvolts shall be recorded to determine compliance with 3.17.14.

4.9.13.1 Test Procedures. -

(a) The test procedures shall be the same as 4.9.1.1 except that 4.9.1.1(b), 4.9.1.1(d), 4.9.1.1.(i), and 4.9.1.1(j) shall not apply.

(b) Connect Hewlett-Packard Signal Generator, Model 606A, or equivalent, to the special pad.

(c) The amount of signal increase in decibels shall not be less than limit specified in 3.17.14.

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4.9.14 Oscillator Radiation (See 3.17.15). - The receiver shall be operated under standard conditions except that no signal shall be applied. A field strength meter, Empire Devices Noise and Field Intensity Meter, or equal, shall be connected to the antenna terminals of the receiver. The frequencies and amplitudes of detectable outputs shall be noted and recorded to determine compliance with 3.17.15. This test shall be conducted at 47.00, 51.00 and 57.00 MHz.

4.9.14.1 Test Procedures. -

(a) Connect a General Radio Output Power Meter, Model 583, or equivalent, to the receiver phone jack. Use 600 ohm load.

(b) Use an adjustable DC Power Supply, Hewlett-Packard Model 721A, or equivalent, to power the receiver under test.

(c) Connect Empire Devices Noise and Field Intensity Meter, or equivalent, to receiver antenna terminals.

(d) Tune Empire Devices meter for oscillator frequency to be checked. Oscillator signal at antenna terminals shall be no greater than limit specified in 3.17.15.

4.9.15 Image Rejection (See 3.17.16). - The receiver shall be operated under standard conditions. The amount of quieting produced by 1 microvolt shall be used as reference. The RF frequency input shall then be set to the image frequencies and the input level increased until the amount of quieting is equal to the reference. The increase in signal level at the image frequencies shall be noted to determine compliance with 3.17.16. The image frequencies which shall be measured are shown in the list.

Image	Frequency
1	Channel Frequency - 910 KHz
2	Channel Frequency - 21.40 MHz
3	Channel Frequency - 20.49 MHz

4.9.15.1 Test Procedures. -

(a) The test procedures shall be the same as 4.9.1.1 except that 4.9.1.1(d), 4.9.1.1(i), and 4.9.1.1(j) shall not apply.

(b) Adjust generator output to level specified in 4.9.15.

(c) The increase in signal level for image frequency being checked shall be no greater than limit specified in 3.17.16.

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4.9.16 Discrimator Characteristics (See 3.17.17). - The receiver shall be operated under standard conditions except that no external signall shall be applied to the receiver input. Under these conditions, apply a 455KHz signal of 50 to 100uv to the first 455 IF amplifier stage from a Hewlett-Packard 606A Signal Generator or equivalent. The output of the discriminator shall be monitored with an RCA Senior Voltohmyst VTVM or equivalent. The input frequency shall then be adjusted in 5KHz steps either side of the center frequency of 455KHz until a frequency range of ±35KHz has been scanned. At each 5KHz point, note and record the voltmeter indication to determine compliance with 3.17.17.

4.9.16.1 Test Procedures. -

(a) Connect a General Radio Output Power Meter, Model 583A, or equivalent, to the receiver phone jack. Use 600ohm load.

(b) Use an adjustable DC Power Supply, Hewlett-Packard Model 721A, or equivalent, to power the receiver under test.

(c) Connect a Hewlett-Packard Model 606A, or equivalent, to filter output TP-1. Adjust generator output to level specified in 4.9.16.

(d) To monitor discriminator output, connect RCA Senior Voltohmyst, or equivalent, to high side of Volume control.

(e) Vary the generator frequency both sides of center frequency and record maximum negative and positive swing on voltohmyst. Record other points noted in 4.9.16. Minimum peak separation and linearity shall be within limits specified in 3.17.17.

4.9.17 Audio Distortion at 12KHz Deviation (See 3.17.18). - The receiver shall be operated under standard test conditions except that the input signal modulation shall be set to 12KHz deviation. The receiver output shall be set at 2.5 times the nominal output level. This level shall be indicated by a General Radio Output Power Meter, Type 583-A, or equivalent. The signal shall then be appropriately coupled to a Hewlett-Packard Distortion Analyzer, Model 330D, or equivalent. Distortion measurement shall then be noted and recorded to determine compliance with 3.17.18.

4.9.17.1 Test Procedures. -

(a) The test procedures shall be the same as 4.9.1.1 except that 4.9.1.1(d), 4.9.1.1(i), and 4.9.1.1(j) shall not apply.

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(b) Connect a Hewlett-Packard Model 330D Distortion Analyzer, or equivalent, to the output power meter terminals.

(c) Adjust generator for 12KHz deviation.

(d) Adjust volume control so that output power meter indicates 2.5 times nominal output level.

(e) Distortion measurement shall be no greater than limit specified in 3.17.18.

4.9.18 Local Oscillator Frequency Accuracy (See 3.17.19). - The receiver shall be operated under standard tests conditions except that no external signal shall be applied. Note the frequency as indicated on the crystal can. With a Hewlett-Packard Counter, Model 524, or equivalent, count the oscillator frequency and record to determine compliance with 3.17.19.

4.9.18.1 Test Procedures. -

(a) Connect a General Radio Output Power Meter, Model 583A, or equivalent, to the receiver under test.

(b) Connect a Hewlett-Packard Model 721A DC Power Supply, or equivalent, to receiver under test.

(c) Use a Hewlett-Packard Model 524D Electronic Counter, or equivalent. Use 2.2pf capacitor to capacity couple electronic counter across crystal.

(d) Frequency counted shall be within limits specified in 3.17.19.

4.9.19 10.245MHz Oscillator Frequency Accuracy (See 3.17.20). - The receiver shall be operated under standard test conditions except that no external signal shall be applied. The oscillator frequency shall then be counted with a Hewlett-Packard Counter, Model 524, or equivalent. Note the indicated frequency and record to determine compliance with 3.17.20.

4.9.19.1 Test Procedures. -

(a) The test procedure shall be the same as 4.9.18.1.

(b) Use 2.2pf capacitor to capacity couple electronic counter across the 10.245MHz crystal.

(c) Frequency counted shall be within limits specified in 3.17.20.

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4.9.20 Frequency Accuracy (See 3.17.21). - The receiver shall be operated under standard test conditions. Monitor discriminator output with RCA Senior Voltohmyst and adjust RF generator for discriminator null. Use Hewlett-Packard Counter, Model 524 or equivalent, to measure frequency of generator. Note frequency and record to determine compliance with 3.17.21. Repeat test at 47 and 57 MHz.

4.9.20.1 Test Procedures. -

(a) The test procedures shall be the same as 4.9.1.1 except that 4.9.1.1(i), and 4.9.1.1(j) shall not apply.

(b) Connect an RCA Senior Voltohmyst to high side of volume control to monitor discriminator null.

(c) Adjust RF generator for a null on RCA Senior Voltohmyst.

(d) Use Hewlett-Packard Counter, Model 524, or equivalent, to measure RF generator frequency.

(3) Frequency measured shall be within limits specified in 3.17.21.

4.10 Standard Test Conditions, Transmitting. - All measurements shall be made at normal room temperature and humidity under the test conditions listed below unless otherwise specified in the individual tests.

Temperature	Room Ambient
Humidity	Room Ambient
Dummy Antenna Load	6.8pf in series with 150ohm resistance (100ohm resistor in series with the 50ohm power meter)
Input Voltage	12.0 volts DC Nominal
Audio Input Voltage	An audio signal is applied to the microphone input terminals through a 1200ohm resistor and to ground.
Frequency	51MHz or 52MHz

4.11 Transmitter Electrical Tests. -

4.11.1 Power Output - Channel 1 (See 3.18.1) - The transmitter radio frequency power output measurements shall be performed as specified in 4.11.1.1. The power output shall be measured at 47, 51 or 52, and 57 megahertz and the results recorded to determine compliance with 3.18.1.

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4.11.1.1 Test Procedures. -

(a) Replace whip antenna with dummy load of 100ohms resistance (non-inductive) in series with 6.8pf capacitance.

(b) Connect dummy load in series with Calorimetric Power Meter, Hewlett-Packard Model 434A, or equal, with an RG-58A/U coaxial cable which is approximately two feet long.

(c) The channel switch CH1 - CH2 shall be in the CH1 position.

(d) Connect 12 VDC to the battery plug from a suitable power supply.

(e) Connect a milliammeter to the power supply leads.

(f) Rotate TONE-VOICE switch to the VOICE position.

(g) Note the power as indicated on the power meter. This indication must be multiplied by a factor of 3 to determine the total power output of the unit. Since the power meter impedance is 50ohms and the dummy load is 100ohms, the meter will only read 1/3 of the total transmitter power output.

(h) Note and record total input current.

4.11.2 Power Output - Channel 2 (See 3.18.2). - The transmitter radio frequency power output measurements shall be performed as specified in 4.11.2.1 and in accordance with the following list. The test results shall be recorded to determine compliance with 3.18.2

Channel 1 Freq. Alignment (MHz)	Channel 2 Freq. Alignment (MHz)
48	47
47	48
47	47
51 or 52	51 or 52
51 or 52	51
51 or 52	53
56	57
57	56
57	57

4.11.2.1 Test Procedures. - Repeat 4.11.1.1 with the exception that the channel switch CH1-CH2 shall be in the CH2 position.

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4.11.3 Tone Oscillator Frequency (See 3.18.3). - The transmitter shall be operated in accordance with 4.11.3.1 with no modulation except the internal tone modulation. The tone frequency shall be measured to determine compliance with 3.18.3.

4.11.3.1 Test Procedures. -

(a) Repeat 4.11.1.1, except that 4.11.1.1(g) and 4.11.1.1(h) shall not apply and with the exception that the TONE-VOICE switch shall be rotated to the TONE position, when test is being performed.

(b) Use a coaxial T-connector at the power meter. From this connector, couple through a 3.3pf capacitance into a coaxial cable which will be connected to a deviation meter.

(c) Connect the low frequency output of the deviation meter into an electronic counter.

(d) Rotate transmitter TONE-VOICE switch to TONE.

(e) Note and record tone oscillator frequency as indicated on the electronic counter.

4.11.4 Modulation Capability (See 3.18.4). - With standard input conditions, the transmitter RF carrier deviation shall meet the requirements of 3.18.4 using the test procedure specified in 4.11.4.1. This test shall be made at the operating frequency of 52 megahertz.

4.11.4.1 Test Procedures. -

(a) Repeat 4.11.3.1 with the exception that 4.11.3.1(c), 4.11.3.1(d) and 4.11.3.1(e) shall not apply.

(b) Remove microphone housing and microphone from transmitter.

(c) Attach 1200ohm resistance test fixture to microphone input terminal.

(d) Connect audio generator signal leads to 1200ohm resistance and ground.

(e) Connect a second pair of audio leads between audio generator output and a vacuum tube voltmeter (VTVM, Hewlett-Packard Model 400D, or equal).

(f) Refer to 3.18.4 for table of audio frequencies and generator output levels at which tests shall be made.

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(g) Rotate TONE-VOICE switch to the VOICE position.

(h) Note and record deviation meter readings at all specified frequencies and respective voltage levels to determine compliance with 3.18.4.

4.11.5 Catching Range (See 3.18.5). - This test shall be performed at 47, 51 or 52, and 57 megahertz in accordance with the test procedures specified in 4.11.5.1. The test results shall comply with 3.18.5.

4.11.5.1 Test Procedures. -

(a) Repeat 4.11.1.1 with the exception that 4.11.1.1(g) and 4.11.1.1(h) shall not apply.

(b) Lightly couple transmitter output into an electronic counter using 3.3 pf capacitance and RG-58A/U coaxial cable.

(c) Use appropriate test lead with 0.040 probe and ground test point TP1.

(d) Rotate TONE-VOICE switch to VOICE.

(e) Adjust VCO trimmer capacitor C27 clockwise until electronic counter reads 180 KHz below the assigned frequency.

(f) Remove grounding lead from test point TP1.

(g) Release TONE-VOICE switch to normal off position.

(h) Rotate TONE-VOICE switch to VOICE position.

(i) Note the frequency on the electronic counter and compare with assigned frequency.

(j) Refer to 3.18.5 to determine compliance.

(k) Repeat steps, 4.11.5.1(c), 4.11.5.1(d) and 4.11.5.1(e).

(l) Adjust VCO trimmer capacitor C27 counterclockwise until electronic counter reads 180 KHz above the assigned frequency.

(m) Repeat steps 4.11.5.1(g) through 4.11.5.1(j).

4.11.6 Frequency Accuracy (See 3.18.6). - The transmitter output frequency shall be measured and recorded at 47, 51 or 52, and 57 megahertz in accordance

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with the test procedures specified in 4.11.6.1. Deviation from the frequency assigned in kilohertz shall be measured and recorded. The frequency accuracy shall comply with 3.18.6.

4.11.6.1 Test Procedures. -

(a) Repeat 4.11.1.1 with exception that 4.11.1.1(g) and 4.11.1.1(h) shall not apply.

(b) Repeat 4.11.5.1(b).

(c) Rotate TONE-VOICE switch to VOICE position.

(d) Note the transmitter output frequency reading on the electronic counter.

(e) Refer to 3.18.6 and determine compliance.

(f) Put channel switch CH1 - CH2 into CH2 position.

(g) Repeat the above steps at 4.11.6.1(c) and 4.11.6.1(d).

4.11.7 VCO Temperature Stability (See 3.18.7). - The test shall be made on Channel 1 at the following VCO frequencies: 47, 51 or 52, and 57 megahertz. The above test shall be repeated on Channel 2 at 52 MHz only. These tests shall be performed in accordance with the test procedures specified in 4.11.7.1. Repeat the test for input voltages of 10.0 and 16.5 VDC. The test results shall comply with 3.18.7.

4.11.7.1 Test Procedures. -

(a) Repeat 4.11.1.1 with the exception that 4.11.1.1(g) and 4.11.1.1(h) shall not apply.

(b) Use appropriate test lead with 0.040 test probe and ground test point TP1.

(c) Place transmitter into a temperature chamber.

(d) Raise temperature in chamber to 131°F.

(e) Lightly couple transmitter output to an electronic counter through a 3.3 pf capacitance and RG-58A/U coaxial cable.

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(f) Note frequency reading on electronic counter and refer to 3.18.7 for compliance.

(g) Adjust power supply voltage to 10.0 VDC.

(h) Repeat 4.11.7.1(f).

(i) Adjust power supply voltage to 16.5 VDC.

(j) Repeat 4.11.7.1(f).

(k) Adjust power supply voltage back to 12.0 VDC and turn off.

(l) Lower temperature in chamber to -40°F.

(m) Turn on power supply.

(n) Repeat steps 4.11.7.1(f) through 4.11.7.1(k).

4.11.8 10.7 MHz Reference Oscillator Stability (See 3.18.8). - The test shall be made with the transmitter operating on Channel 1 at 52 MHz and at input voltages of 10.0 VDC, 12.0 VDC and 16.5 VDC. The test shall be performed in accordance with the test procedures specified in 4.11.8.1. The test results shall comply with 3.18.8.

4.11.8.1 Test Procedures. -

(a) Repeat 4.11.1.1 with the exception that 4.11.1.1(g) and 4.11.1.1(h) shall not apply.

(b) Couple the reference oscillator buffer amplifier output, which is found at the funnelet at side of T6, to an electronic counter through a 3.3 pf capacitance and RG-58A/U coaxial cable.

(c) Place transmitter into a temperature chamber.

(d) Raise temperature in chamber to 131°F.

(e) Note frequency reading on electronic counter and refer to 3.18.8 for compliance.

(f) Adjust power supply voltage to 10.0 VDC.

(g) Repeat 4.11.8.1(e)

(h) Adjust power supply voltage to 16.5 VDC.

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(i) Repeat 4.11.8.1(e).

(j) Adjust power supply voltage to 12.0 VDC and turn off.

(k) Lower temperature in chamber to -40°F.

(l) Turn on power supply.

(m) Repeat steps 4.11.8.1(e) through 4.11.8.1(j).

4.11.9 Input Voltage Range (See 3.18.9). - To determine compliance with 3.18.9, the tests listed below shall be conducted at low and high input voltages in accordance with the following test procedure.

(a) Power Output (Channel 1) (4.11.1).

(b) Frequency Accuracy (4.11.6).

(c) Regulated Voltage (4.11.20).

(d) VCO Temperature Stability (4.11.7).

(e) 10.7 Reference Oscillator (4.11.8).

(f) Tone Squelch Generator Frequency (4.11.21).

4.11.9.1 Test Procedures. -

(a) Repeat 4.11.1.1 with the exception that 4.11.1.1(d) shall not apply.

(b) Adjust power supply output voltage at 10.0 VDC.

(c) Lightly couple transmitter output to an electronic counter through 3.3 pf capacitance and RG-58A/U coaxial cable.

(d) Connect VTVM (Hewlett-Packard 410B or equal) to voltage regulator output which is found at eyelet-end of L6.

(e) Rotate TONE-VOICE switch to VOICE position.

(f) Using a T-connector, couple the power meter through a 3.3 pf capacitor and a coaxial cable to a deviation meter.

(g) Connect the low frequency output of the deviation meter to an electronic counter.

(h) Note power output and refer to 3.18.1 for compliance.

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(i) Note output frequency reading on electronic counter and refer to 3.18.6 for compliance.

(j) Note regulated voltage on VTVM and refer to 3.18.20 for compliance.

(k) Note tone squelch generator reading on low frequency electronic counter and refer to 3.18.21 for compliance.

(l) Release TONE-VOICE switch to normal OFF position.

(m) Adjust power supply output voltage to 16.5 VDC.

(n) Repeat steps 4.11.9.1(e) through 4.11.9.1(l).

4.11.10 System Efficiency (See 3.18.10). - The transmitter shall be operated under nominal input voltage conditions. The input and output powers shall be recorded. The ratio of output-to-input power shall comply with 3.18.10. The following test procedure shall be used.

4.11.10.1 - Repeat 4.11.1.1(a) through 4.11.1.1(h) and note input and output power.

4.11.10.2 - Refer to 3.18.10 for compliance.

4.11.11 Frequency Range (See 3.18.11). - The transmitter shall be aligned at 47 MHz on Channel 1. The transmitter shall comply with 3.18.1 and 3.18.6. Realign transmitter at 57 MHz. Transmitter again shall comply with 3.18.1 and 3.18.6.

4.11.11.1 Test Procedures. -

(a) Insert a 36.3 MHz crystal into CH1 crystal holder.

(b) Repeat 4.11.1.1(a) through 4.11.1.1(f).

(c) Completely align transmitter to 47.0 MHz.

(d) Note power output on power meter and refer to 3.18.1 for compliance.

(e) Lightly couple transmitter output to an electronic counter through a 3.3 pf capacitance and RG-58A/U coaxial cable.

(f) Note frequency reading on electronic counter and refer to 3.18.6 for compliance.

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(g) Remove 36.3 MHz crystal from CH1 holder and insert a 46.3 MHz crystal.

(h) Realign transmitter to 57 MHz.

(i) Repeat steps 4.11.11.1(d) through 4.11.11.1(f).

4.11.12 Audio Input Impedance (See 3.18.12). - The input circuit impedance shall be measured at frequencies in the range of 300 to 3000 Hz and shall meet the requirements as specified in 3.18.12.

4.11.12.1 Test Procedures. -

(a) Repeat 4.11.1.1 with the exception that 4.11.1.1(g) and 4.11.1.1(h) shall not apply.

(b) Remove microphone housing and microphone from transmitter.

(c) Connect a General Radio decade box Type No. 1423-L or equal to the audio input.

(d) Connect an audio signal generator with an output impedance of 100 ohms or less between the second terminals on decade box and ground on transmitter such that the decade box will be in series with the generator and the audio circuit.

(e) Connect #1 VTVM (Hewlett-Packard Model 400D or equal) to audio generator output and ground.

(f) Set audio generator output to read 6 mv on VTVM and retain this level throughout the test.

(g) Connect #2 VTVM (Hewlett-Packard Model 400D or equal) between the audio input circuit and ground.

(h) Set the audio generator at 300 Hz.

(i) Recheck 4.11.12.1(f) and readjust if necessary.

(j) Vary decade box unit until VTVM #2 reads one-half of VTVM #1.

(k) The decade box setting is the audio input impedance. Refer to 3.18.12 for compliance.

(l) Set audio generator to 1000 Hz.

(m) Repeat steps 4.11.12.1(h) through 4.11.12.1(j).

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(n) Set audio generator to 3000 Hz.

(o) Repeat steps 4.11.12.1(h) through 4.11.12.1(i).

4.11.13 Deviation Limiting (See 3.18.13). - The deviation of the transmitter shall be set at 8 KHz using an input modulating frequency of 1000 Hz. The input voltage level shall be recorded. The modulating input voltage shall then be increased by 20 db and the deviation measured and recorded. Maximum deviation shall comply with 3.18.13.

4.11.13.1 Test Procedures. -

(a) Repeat 4.11.1.1 with the exception that 4.11.1.1(g) and 4.11.1.1(h) shall not apply.

(b) Remove microphone housing and microphone.

(c) Attach 1200 ohm resistance test fixture to microphone input terminal.

(d) Connect audio generator across 1200 ohm resistance and ground.

(e) Connect a second pair of audio leads between audio generator output and a VTVM (Hewlett-Packard Model 400D or equal).

(f) Use a coaxial T-connector at the power meter. From this connector, couple through a 3.3 pf capacitance into a coaxial cable which connects to a deviation meter.

(g) Set audio generator output level to 5.6 mv as indicated by the VTVM. This level shall be sufficient to produce 8 KHz (+4 KHz, -2 KHz) deviation.

(h) Increase the audio generator output signal level by 20 db as indicated on the VTVM decibels scales.

(i) Note the deviation meter reading and refer to 3.18.13 to determine compliance.

4.11.14 Spurious Radiations (See 3.18.14). - The transmitter shall be operated under the standard test conditions except that no modulation shall be applied. With the transmitter tuned to 47, 51 or 52, or 57 MHz, the field intensity meter shall be tuned throughout the 20 to 400 MHz range. At points of response, the attenuation shall be measured to determine compliance with 3.18.14.

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4.11.14.1 Test Procedures. -

(a) Repeat 4.11.1.1 with the exception that 4.11.1.1(h) shall not apply.

(b) Connect an RG-58A/U coaxial cable, with a length of approximately 2 feet, across the power meter.

(c) Connect the other end of the coaxial cable to coaxial attenuator pads which will have a total attenuation of 30 db.

(d) Connect the attenuator pads to the Empire Devices NC-105 Noise and Field Intensity Meter or equal switch unit SIGNAL INPUT receptacle.

(e) Sweep the Empire Devices NC-105 through the 20 to 400 MHz range and record all spurious responses which are less than 80 db below the radiated transmitter frequency. Refer to 3.18.14 to determine compliance.

(f) Switch channel selector to CH2 position.

(g) Remove 10 db from coaxial attenuator pads.

(h) Repeat 4.11.14.1(e).

(i) Switch channel selector to CH1 position.

4.11.15 Harmonic Radiations (See 3.18.15). - The transmitter shall be operated under the standard test conditions except that no modulation shall be applied. With the transmitter tuned to 47, 51 or 52 or 57 MHz, the field intensity meter shall be tuned throughout the 94 to 400 MHz tuning range. At points of response, the attenuation shall be measured to determine compliance with 3.18.15.

4.11.15.1 Test Procedures. -

(a) Repeat 4.11.14.1(a) through 4.11.14.1(d).

(b) Sweep the Empire Devices NC-105 Noise and Field Intensity Meter or equal through the 94 to 400 MHz range and record all harmonic radiation suppression levels. Refer to 3.18.15 to determine compliance.

(c) Repeat 4.11.14.1(f) and 4.11.14.1(g).

(d) Repeat 4.11.15.1(b).

(e) Switch channel selector to CH1 position.

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4.11.16 AM Modulation (See 3.18.16). - The transmitter shall be operated under the standard test conditions with external modulation applied at 1000 Hz and sufficient amplitude to cause maximum carrier deviation. The ratio of the detected peak AC carrier to the detected DC voltage shall then be noted to determine compliance with 3.18.16.

4.11.16.1 Test Procedures. -

(a) Repeat 4.11.13.1.

(b) Disconnect transmitter signal from deviation meter.

(c) Connect transmitter signal to a Hewlett-Packard 420B RF crystal detector, or equal.

(d) Connect the crystal detector output to the Tektronic oscilloscope Model 531 high gain amplifier plug-in unit, or equal.

(e) With the vertical sweep input selector switch placed in the AC position, observe and record the detected peak modulating audio signal.

(f) The vertical sweep input selector switch is placed in the DC position and the detected DC voltage is observed and recorded.

4.11.17 Antenna Loading (See 3.18.17). - The transmitter shall be equipped with its appropriate antenna and operated under standard conditions except that no modulation be applied. As a measure for determining compliance with 3.18.17, the loading coil shall be capable of resonating with the appropriate antenna at 47 and 57 MHz.

4.11.17.1 Test Procedures. -

(a) The transmitter AN/PRT-4( ) shall be fully equipped with its appropriate 24-inch whip antenna and battery.

(b) With the antenna fully extended and the Voice-Tone switch in the Voice position, place the transmitter antenna near a field strength meter to obtain an adequate field strength reading.

(c) Adjust antenna loading coil L12 until a definite field strength peak is obtained.

4.11.18 Audio Frequency Response (See 3.18.18). - The transmitter shall be operated under standard conditions with external modulation applied through 1200 ohms to the microphone input circuit. The audio level shall be sufficient

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to cause 8 KHz carrier deviation. The transmitter output shall be adequately attenuated and fed into a deviation monitor, Marconi Instruments PF791D, or equal. The audio frequencies at which measurement shall be made are 200, 400, 800, 1600, and 3200 Hz. The audio frequency response is determined by the decrease or increase of the audio input in db required to maintain ±8 KHz deviation at the various audio frequencies. The audio input variations shall be recorded to determine compliance with 3.18.18.

4.11.18.1 Test Procedures. -

(a) Repeat 4.11.4.1 with the exception that 4.11.4.1(f) and 4.11.4.1(h) shall not apply.

(b) The audio generator output signal is adjusted to maintain ±8 KHz deviation and recorded for each of the audio test frequencies specified in 4.11.18.

4.11.19 System Distortion (See 3.18.19). - This test shall be made with a companion approved AN/PRR-9( ) receiver. With both receiver and transmitter operating under appropriate standard test conditions in accordance with the test procedure specified in 4.11.19.1, the distortion measured at 1000 Hz shall comply with the requirements of 3.18.19.

4.11.19.1 Test Procedures. -

(a) Repeat 4.11.18.1 for transmitter AN/PRT-4( ) set up.

(b) Repeat 4.9.1.1 and 4.9.2.1 with the exception of 4.9.1.1(b) and 4.9.2.1(b) for receiver AN/PRR-9( ) set up.

(c) Replace Measurements Corporation Signal Generator, Model 560, with an attenuated transmitter AN/PRT-4( ) signal level, which is at the appropriate receiver frequency, between 50 uv to 5000 uv.

(d) Connect input terminals of 150 Hz Twin T network to power meter terminals.

(e) Connect Hewlett-Packard Model 330D Distortion Analyzer, or equivalent, to the output of the Twin-T network.

(f) Measure the system distortion by observing the distortion as read on the distortion analyzer.

4.11.20 Regulated Voltage (See 3.18.20). - The regulated voltage measurements shall be performed when the regulated voltage output is properly loaded

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cause variations in power output.

with 30 ma ±20 ma. The regulator voltage load is measured as specified in test procedures specified in 4.11.20.1. The regulated voltage is measured and recorded with input voltage variations from 10.0 VDC to 16.5 VDC. The recorded results shall comply with 3.18.20.

4.11.20.1 Test Procedures. -

(a) The regulated voltage shall be measured by connecting a DC VTVM to the top (adjusting end) of trimmer capacitor C27 (VCO trimmer capacitor).

(b) The regulated voltage load (current drain) shall be measured by adjusting the trimmer capacitors C43 and C45 for minimum input current. The total remaining input current shall essentially be the regulated voltage load current.

4.11.21 Tone Squelch Generator Frequency (See 3.18.21). - The transmitter shall be operated under the standard conditions with no modulation except the 150 Hz tone modulation. The tone frequency shall be measured to determine compliance with 3.18.21.

4.11.21.1 Test Procedures. -

(a) Repeat 4.11.3.1 except that 4.11.3.1(d) and 4.11.3.1(e) shall not apply, and in 4.11.3.1(a), the TONE-VOICE switch shall be rotated to the VOICE position when the test is being performed.

(b) Without voice modulation, note and record, tone squelch generator frequency, as indicated on the electronic counter.

4.11.22 Tone Squelch Generator Modulation (See 3.18.22). - The transmitter shall be operated under standard conditions with no modulation except the internal 150 Hz tone modulation. The transmitter deviation shall be measured to determine compliance with 3.18.22.

4.11.22.1 Test Procedures. -

(a) Repeat 4.11.1.1 except that 4.11.1.1(g) and 4.11.1.1(h) shall not apply.

(b) Use a coaxial T-connector at the power meter. From this connector, couple through a 3.3 pf capacitor and a coaxial cable, to a deviation meter.

(c) Without voice modulation, note and record deviation reading and refer to 3.18.22 for compliance.

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4.11.23 Tone Squelch Generator Temperature Stability (See 3.18.23). - The frequency of the tone squelch generator shall be measured at room temperature. The ambient temperature shall then be raised to 131°F, and held at that temperature for one hour, and the frequency measured. The test shall be repeated for an ambient temperature of -40°F, and at the end of a one hour stabilization period at the low temperature limit, shall be recorded to establish compliance with 3.18.23.

4.11.23.1 Test Procedures. -

(a) Repeat 4.11.21.1

(b) Place transmitter in temperature box and reduce ambient temperature to -40°F.

(c) After one hour, repeat 4.11.21.1.

(d) Raise ambient temperature to +131°F.

(e) After one hour repeat 4.11.21.1.

4.11.24 Tone Squelch Generator Distortion (See 3.18.19). - The transmitter shall be operated under standard conditions with no modulation except the internal 150 Hz tone squelch modulation.

4.11.24.1 Test Procedures. -

(a) Repeat 4.11.3.1 steps (a) and (b), except that the TONE-VOICE switch shall be rotated to the VOICE position when the test is being performed.

(b) Connect the low frequency output of the deviation meter into a distortion analyzer.

(c) Note and record the percent distortion at 150 Hz, to determine compliance with 3.18.19.

4.12 Inspection for Interchangeability. - The mechanical dimensions and electrical performance required by the applicable mechanical and electrical drawings shall be gaged or measured to determine conformance to the physical and functional interchangeability requirements of 3.19. When a dimension or performance requirement is not within specified or design limits, it shall be considered a major defect.

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4.13 Reliability Test (See 3.20.- Receivers and transmitters, selected from alternate monthly lots that have passed Group A and Group B (4.5.1 and 4.5.2), and Group C, subgroups I and II (4.5.3) shall be subjected to reliability testing in accordance with the Reliability Test Plan for Squad Radio (Appendix A). The month's production represented by the samples therefrom under reliability test shall not be shipped until the samples have successfully completed the reliability test. Units subjected to the reliability test shall be refurbished and retested to the satisfaction of the Government representative and will be submitted as part of the production equipment.

4.14 Operational Inspection (See 3.23).- Immediately prior to packaging, a receiver listening test shall be made with the receiver operated under standard test conditions except that the squelch shall be on. With the transmitter operated under standard test conditions, the power output shall be measured in Channel 1 and Channel 2. A standard receiver shall be tuned to the transmitter frequency and a listening test shall be made when transmitting voice modulation. The test shall be performed on both Channel 1 and Channel 2.

4.15 Acceptance Inspection of Preparation for Delivery.- Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.16 Rough Handling Test (Preparation for Delivery).- When rough handling test in accordance with MIL-P-116 is required by the contract (See 6.2), the operational inspection (See 4.14) of the receiver and transmitter shall be conducted to determine freedom from operational malfunction caused by rough handling.

4.17 Visual and Mechanical Inspection.- The units shall be examined for the applicable defects listed in MIL-STD-252.

4.18 Fungus Test.- The equipment (including accessories) shall be tested in accordance with Method 508, Procedure 1, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required.

4.19 Salt Fog Test.- The equipment (including accessories) in its normal operating position shall be tested in accordance with Method 509, Procedure 1, of MIL-STD-810. The equipment shall not be opened. Operation is not required. No rust shall be imposed on the test item by the test facility.

5. PREPARATION FOR DELIVERY

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5.1 Preservation and Packaging. - Preservation and packaging shall be level A or C as specified (See 6.2(c)).

5.1.1 Level A. -

5.1.1.1 Cleaning. - Receiving Set, Radio AN/PRR-9( ) or Transmitting Set, Radio AN/PRT-4( ) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Receiving Set, Radio AN/PRR-9( ) or Transmitting Set, AN/PRT-4( ) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation Application. - None required.

5.1.1.4 Unit Packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical Literature. - Each technical literature shall be packaged Method 1C-1.

5.1.1.4.2 Receiving Set, Radio AN/PRR-9( ). - Each set shall be packaged Method III, as follows:

5.1.1.4.2.1 Antenna Assembly AS-1998( )/PRR-9. - Cushion each antenna by wrapping in paperboard conforming to PPP-P-291, Type III. Secure the wrap with tape conforming to PPP-T-45, Type III.

5.1.1.4.2.2 Headset, Electrical H-264( )/PRR-9. - Each headset shall be cushioned in accordance with the procedure specified in 5.1.1.4.2.1.

5.1.1.4.2.3 Lanyard Assembly. - Each assembly shall be wound into a coil of proportionate dimensions and placed within a bag conforming to UU-B-36.

5.1.1.4.2.4 Slide, Harness. - Each slide shall be placed within a bag specified in 5.1.1.4.2.3.

5.1.1.4.2.5 Receiver Assembly. - Each assembly shall be cushioned in accordance with the procedure specified in 5.1.1.4.2.1.

5.1.1.4.2.6 Consolidated Package. - Consolidate the items, packaged as specified in 5.1.1.4.2.1 through 5.1.1.4.2.5, within a close-fitting fiberboard box conforming to PPP-B-636, W6c. Fill all voids with paperboard conforming to PPP-P-291, to prevent movement. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

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5.1.1.4.3 Transmitting Set, Radio AN/PRT-4( ). - Each set shall be packaged Method III, as follows:

5.1.1.4.3.1 Lanyard Assembly. - Each assembly shall be packaged as specified in 5.1.1.4.2.3.

5.1.1.4.3.2 Slide, Harness. - Each slide shall be packaged as specified in 5.1.1.4.2.4.

5.1.1.4.3.3 Transmitter Assembly. - Each assembly, including the attached Antenna Assembly AS-1999( )/PRT-4( ), shall be packaged as specified in 5.1.1.4.2.5.

5.1.1.4.3.4 Consolidated Package. - Consolidate the items, packaged as specified in 5.1.1.4.3.1 through 5.1.1.4.3.3, within a close-fitting fiberboard box conforming to PPP-B-636, W6c. Fill all voids with paperboard conforming to PPP-P-291, to prevent movement. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C. -

5.1.2.1 Reveiving Set, Radio AN/PRR-9( ). - Each set shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.1.2.2 Transmitting Set, Radio AN/PRT-4( ). - Each set shall be preserved and packaged as specified in 5.1.2.1.

5.2 Packing. - Packing shall be level A, B or C as specified (See 6.2(c)).

5.2.1 Level A. -

5.2.1.1 A quantity of Receiving Set, Radio AN/PRR-9( ) or Transmitting Set, Radio AN/PRT-4( ), packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, Type CF, class weather-resistant, special requirements use. Box closure shall be in accordance with the appendix to the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

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5.2.1.2 Palletized Load. - A quantity of containers, packed as specified in 5.2.1.1 shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less Than Palletized Load. - When quantities per destination are less than a pallet load, containers specified in 5.2.1.1, shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix to the box specification. No further packing shall be required.

5.2.2 Level B. - A quantity of Receiver Set, Radio AN/PRR-9( ) or Transmitting Set, Radio AN/PRT-4( ), packaged as specified in 5.1, shall be packed as specified in 5.2.1.

5.2.3 Level C. - A quantity of Receiver Set, Radio AN/PRR-9( ) or Transmitting Set, Radio AN/PRT-4( ), packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that the pallets, fiberboard boxes and fiberboard caps shall be of the type, size and kind commonly used for the purpose and shall comply with rules and regulations of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. - The Squad Radio Set, consisting of Radio Receiver AN/PRR-9( ) and Radio Transmitter AN/PRT-4( ) is intended to provide short range voice communications (up to 1 mile) for combat troops. A second transmitter channel is provided for very short range voice communications (up to 500 yards) for combat troops.

6.2 Ordering Data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Levels of preservation, packaging, packing and marking (See Section 5).

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(d) The specific paragraphs of Section 5 which are applicable to the particular procurement.

(e) When the rough handling and functional tests are required (See 4.16).

(f) Place of final inspection.

(g) Technical literature required. (3.21).

(h) Submission of the statement of treatment referenced in 3.24.5, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Color.- The color chip furnished by the procuring agency (3.8) will be color chip Number X-24087, and may be obtained upon request to Commanding General, US Army Electronics Command, ATTN: AMSEL-PP/P, 225 South Eighteenth Street, Philadelphia, Pennsylvania 19103.

6.4 Definition of Level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.5 Group C Inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2).

6.6 Location of Operational Inspection.- It is desirable that the operational inspection (4.14) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.7 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example; AN/PRT-4W. The supplier should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

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6.8 Verification Inspection. - Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this document and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the supplier's quality control system and the quality history of the product.

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APPENDIX A

RELIABILITY TEST PLAN FOR SQUAD RADIO

A10.0 SCOPE - This test plan defines the methods and procedures to be used to perform the Reliability Test.

A20.0 APPLICABLE DOCUMENTS - The following documents of the issue in effect on date of Invitation for Bids, or Request for Proposal, form a part of this appendix to the extent specified herein:

MIL-STD-781 Reliability Tests: Exponential Distribution
MIL-STD-785 Requirements for Reliability Program (for Systems and Equipments)

A30.0 Reliability Requirement (MTBF). The Receiving Set, Radio AN/PRR-9( ) and Transmitting Set, Radio AN/PRT-4( ) shall be capable of providing a specified mean-time-between-failure (MTBF) of 1000 and 850 hours, respectively, while operating in the following manner:

a. A temperature cycling environment from -40°C to +55°C.

b. The AN/PRR-9( ) shall operate at a duty cycle of 100%.

c. The AN/PRT-4( ) shall operate key down 25 minutes for each 30 minutes of on time.

A30.1 Reliability Program. The supplier shall conduct a reliability program to insure that the design reliability is maintained throughout production. The program shall include, but not be limited to, the following tasks described in MIL-STD-785:

a. Defective or inadequate parts/specifications.

b. Critical items.

c. Supplier and subcontractor reliability programs.

d. Effects of storage, shelf-life, packaging, transportation, handling and maintenance.

e. Manufacturing controls and monitoring.

f. Failure data collection, analysis and corrective action.

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A40.0 Reliability Testing. MIL-STD-781 shall apply in its entirety unless otherwise specified herein.

A40.1 Reliability Test Conditions. The following conditions shall apply during the reliability test program:

A40.1.1 Demonstration Environment. All reliability demonstrations shall be conducted under the environment of test level "E", 4.1.6 of MIL-STD-781, except that the low temperature shall be modified to -40°C. The standard method of temperature cycling, 5.2.3.1 of MIL-STD-781, shall be imposed. The rate of temperature change shall be not less than 5°C/minute.

A40.1.2 Duty Cycle. During the on time portion of the reliability test cycle, the AN/PRR-9( ) shall operate 100% of the time. The AN/PRT-4( ) shall operate key down for 25 minutes out of every 30 minutes on time.

A40.1.3 Parameters to be Measured. The following parameters are to be measured and recorded daily. Measurements shall be taken above and below as well as at nominal voltages. Vibration shall not be avoided during measurements. Testing shall be in accordance with the referenced test paragraphs below:

	Rec'r Para	Test Para
Receiver AN/PRR-9( )		
Sensitivity	3.17.1	4.9.1
Distortion	3.17.2	4.9.2
Audio Output Power	3.17.3	4.9.3
Squelch Sensitivity	3.17.6	4.9.6
Frequency Accuracy	3.17.21	4.9.20
Transmitter AN/PRT-4( )		
Power Output (Channel 1)	3.18.1	4.11.1
Modulation Capability	3.18.4	4.11.4
Catching Range	3.18.5	4.11.5
Frequency Accuracy	3.18.6	4.11.6

A40.1.4 Failure Definition. Failure is defined as the inability of the equipment to perform the required function---this includes any event which causes departure from performance as required by the equipment specification (See A40.1.3). In addition, 5.5.1 of MIL-STD-781 shall apply.

A40.1.5 Voltage Cycling. The transmitter and receiver voltages shall be cycled as follows: 10.0 volts and 4.0 volts 25% of the "on time", 12.0 volts and 5.0 volts 50% of the "on time", and 16.5 volts and 6.0 volts 25% of the "on time".

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A40.2 Reliability Qualification Demonstration. A minimum of 10 each receivers and transmitters shall be subjected to reliability testing after other preproduction tests have been completed (See 5.1.2 and 5.4.1 of MIL-STD-781). These samples shall be subjected to the accept/reject criteria of Test Plan IV, 4.2.8.4 of MIL-STD-781. The MTBF (specified) shall be defined as 1000 and 850 hours, respectively.

A40.3 Reliability Production Phase Demonstration. Subsequent to successful completion of the qualification demonstration, a sample of 10 receivers and 10 transmitters shall be randomly selected from alternate monthly lots. The production sampling tests shall commence with the first lot available after successful completion of the reliability qualification test. The samples shall be subjected to the accept/reject criteria of Test Plan IVA, 4.2.8.4a of MIL-STD-781. The MTBF (specified) shall be defined as 1000 and 850 hours, respectively.

A40.4 Reliability Test Reports. Reliability test reports shall be submitted in accordance with MIL-STD-781.

A40.5 Corrective Action. Corrective action shall be in accordance with MIL-STD-781. In addition, the supplier shall immediately investigate and report details of the proposed corrective action on all units of product which were manufactured under the same conditions and which the government considers subject to the same failure.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-R-55[illegible]A(EL), RADIO, SQUAD, RECEIVING AND TRANSMITTING (UNITS OF)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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